

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS: BTech / IMSc (PHY)
BRANCH: BT/CIVIL/CHEMICAL/MECH/PIE/FT/PHYSICS

SEMESTER : I
SESSION : MO/2024

SUBJECT: PH24101/PH113 PHYSICS

TIME: 3 Hours

FULL MARKS: 50

INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
 2. Attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Before attempting the question paper, be sure that you have got the correct question paper.
 5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.
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Q.1(a)	What is interference?	[2]	1 1
Q.1(b)	Light of wavelength $\lambda = 630$ nanometer (nm) falls normally on a thin wedge-shaped film formed by a material with refractive index $n = 1.8$. There are ten bright and nine dark fringes over the length of the film. By how much does the film thickness change over this length?	[3]	1 2
Q.1(c)	Briefly outline (few sentences only) the phenomena of single slit diffraction. Draw the intensity pattern with suitable labels.	[5]	1 3
Q.2(a)	Discuss any two Maxwell's equations in free space (charges and currents may be present).	[4]	2 2
Q.2(b)	Show that the normal component of the magnetic flux density (B) is continuous at a boundary between two media. At the same time, the normal component of magnetic field intensity (H) is discontinuous.	[6]	2 3
Q.3(a)	Show that relativistic length squared, $ds^2 = c^2t^2 - x^2 - y^2 - z^2$, remains conserved under Lorentz transformations.	[5]	3 3
Q.3(b)	Two spaceships approach each other, each moving with the same speed as measured by a stationary observer on the Earth. Their relative speed is $0.50c$, Determine the velocities of each spaceship as measured by the stationary observer on Earth.	[5]	3 3
Q.4(a)	Write down the expression for change in wavelength of the x-ray incident on a metal target (Compton effect). Determine the wavelength of the outgoing x-ray when scattered at an angle of 10 degrees. The wavelength of the incident x-ray was 10 pm.	[4]	4 2
Q.4(b)	Write down the time independent Schrodinger equation for a quantum particle in a one-dimensional box and obtain possible energies.	[6]	3
What will be the energies of the first and second excited states if the ground state energy is given to be 11.6 eV (electronvolts)?			
Q.5(a)	What are nuclear magic numbers ? Classify the following nuclei as magic, non-magic or doubly magic: ${}^4_2\text{He}$ (Helium-4), ${}^{59}_{27}\text{Co}$ (Cobalt-59), ${}^{12}_6\text{C}$ (Carbon-12).	[5]	5 3
Q.5(b)	Discuss the basic working principle of a laser using the example of He-Ne laser. Mention any two applications of lasers.	[5]	5 2

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